PRACTICE SHEET

1. Sum of all odd position terms of the expansion $(1 + x)^8$?
(a) 256 (b) 512
(c) 1024 (d) 128

2. What are the last two digits of the number 9^{200} ?
(a) 19 (b) 21
(c) 41 (d) 01

3. For any positive integer n, if $4^n - 3n$ is divided by 9, then what is the remainder?
(a) 8 (b) 6
(c) 4 (d) 1

4. What is the coefficient of x^5 in the expansion $(1 - 2x + 3x^2 - 4x^3 + ... + \infty)^{-5}$?
(a) $(10!) / (5!)^2$ (b) 5^{-5}
(c) 55 (d) $10!/\{6!)(4!)\}$

5. What is the middle term in the expansion of $\left(\frac{x\sqrt{y}}{3} - \frac{3}{y\sqrt{x}}\right)^{12}$
(a) $C(12, 7)\, x^3 y^{-3}$ (b) $C(12, 6)\, x^{-3} y^3$
(c) C(12, 7) x–3 y3 (d) C(12, 6) x3 y–3

6. If x^4 occurs in the r^{th} term in the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$, then what is the value of r?
(a) 4 (b) 8
(c) 9 (d) 10

7. After simplification, what is the number of terms in the expansion of $[(3x + y)^5]^4 - [3x - y)^4]^5$?
(a) 4 (b) 5
(c) 10 (d) 11

8. What is the coefficient of $x^3 y^4$ in $(2x + 3y^2)^5$?
(a) 240 (b) 360
(c) 720 (d) 1080

9. What is the last digit of $3^{3^{4n}} + 1$, where n is a natural number?
(a) 2 (b) 7
(c) 8 (d) None of these

10. If t_r is the r^{th} term in the expansion of $(1 + x)^{101}$, then what is the ratio $\frac{t_{20}}{t_{19}}$ equal to
(a) $\frac{20x}{19}$ (b) 83x
(c) 19x (d) $\frac{83x}{19}$

11. What is the value of
${}^8C_0 - {}^8C_1 + {}^8C_2 - {}^8C_3 + {}^8C_4 - {}^8C_5 + {}^8C_6 - {}^8C_7 + {}^8C_8$
(a) 0 (b) 1
(c) 2 (d) 2^8

12. What is the term independent of x in the expansion of $\left(1 + x + 2x^3\right)\left(\frac{3x^2}{3} - \frac{1}{3x}\right)^9$
(a) 1/3
(b) 17/54
(c) ¼
(d) No such term exists in the expansion

13. What is the coefficient of x^4 in the expansion of $(1 + 2x + 3x^2 + 4x^3 + ...)^{1/2}$?
(a) 1/4 (b) 1/16
(c) 1 (d) 1/28

14. Consider the following statements.
I. The coefficient of the middle term in the expansion of $(1 + x)^8$ is equal to the middle term of $\left(x + \frac{1}{x}\right)^8$
II. The coefficient of the middle term in the expansion of $(1 + x)^8$ is less than the coefficient of the fifth term in the expansion of $(1 + x)^7$.
Which of the above statements is/are correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

15. How many zeroes are there in 80!
(a) 16 (b) 18
(c) 19 (d) 20

16. Find the remainder $\frac{16^{200} + 5}{15}$
(a) 6 (b) 7
(c) 10 (d) 12

17. What is the sum of odd place coefficients in the binomial expansion of $(a + b)^n$
(a) 2^n (b) 2^{n+1}
(c) 2^{n-1} (d) 2

18. The coefficient of x^7 in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^8$ is
(a) ${}^8C_3 \frac{a^3}{b^5}$ (b) ${}^8C_3 \frac{a^5}{b^3}$
(c) ${}^8C_4 \frac{a^5}{b^3}$ (d) None of these

19. The coefficient of x^{-7} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^8$
(a) $(-1)^5\, {}^8C_5 \frac{a^3}{b^5}$ (b) $(-1)^5\, {}^8C_5 \frac{a^5}{b^3}$
(c) ${}^8C_3 \frac{a^5}{b^3}$ (d) None of these

20. What is the coefficient of x^4 in the expansion of $\left(\frac{1-x}{1+x}\right)^2$?
(a) –16 (b) 16
(c) 8 (d) –8

21. What is the sum of the coefficient of all the terms in the expansion of $(45x - 49)^4$?
(a) –256 (b) –100
(c) 100 (d) 256

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22. What is the number of terms in the expansion of $(a + b + c)^n$, $n \in N$?

(a) $n + 1$ (b) $n + 2$ (c) $n(n + 1)$ (d) $\frac{(n+1)(n+2)}{2}$

ANSWER KEY

1.	d	2.	d	3.	d	4.	a	5.	d	6.	c	7.	c	8.	c	9.	d	10.	d
11.	a	12.	b	13.	c	14.	a	15.	c	16.	a	17.	c	18.	b	19.	a	20.	b
21.	d	22.	d																

Solutions

Sol.1. (d)

Sum of all odd position terms
$= 2^{n-1} = 2^7 = 128$

Sol.2. (d)

Using binomial theorem $9^{200} = (1 + 8)^{2000}$

$= 1 + 8.200 + \frac{200 \times 199}{2!} \times 8^2 + ...$

$= 1 + 1600 + 1273600 + ...$

From above, it is clear that the last two digits of the number 9^{200} are 01

Sol.3. (d)

Using binomial theorem.

$4^n - 3n = (1 + 3)n - 3n$

$1 + n.3 + \frac{n(n-1)}{2!} 3^2 + ... - 3n$

$= 1 + \frac{n(n-1)}{2!}.3^2 + \frac{n(n-1)(n-2)}{3!}.3^3 + ...$

$\Rightarrow 4^n - 3n = 9\left\{\frac{n(n-1)}{2!} + ...\right\} + 1$

Thus, when $4n - 3n$ is divided by 9, the remainder is 1.

Sol.4. (a)

$1 - 2x + 3x^2 - 4x^3 + ...$

$= (1 + x)^{-2}$, so, $(1 - 2x + 3x^2 - 4x^3 + ...\infty)^{-5}$

$= ((1+x)^{-2})^{-5} = (1+x)^{10} \Rightarrow T_{r+1} = {}^{10}C_r X^r$

Putting r = 5, coefficient of $x^5 = {}^{10}C_5 = \frac{10!}{5!5!} = \frac{10!}{(5!)^2}$

Sol.5. (d)

In the expansion of $\left(\frac{x\sqrt{y}}{3} - \frac{3}{y\sqrt{x}}\right)^{12}$, then middle term is $\frac{12}{2} + 1 = 7^{th}$ term, $(r + 1)_{th}$ term,

$T_{r+1} = {}^{12}C_r \left[\frac{x\sqrt{y}}{3}\right]^{12-r} \left(-\frac{3}{y\sqrt{x}}\right)^r$

$\therefore T_7 = T_{6+1} = {}^{12}C_6 \left(\frac{x\sqrt{y}}{3}\right)^6 \left(-\frac{3}{y\sqrt{x}}\right)^6$

$= {}^{12}C_6 \frac{x^6 y^3}{y^6 x^3} = {}^{12}C_6 x^3 y^{-3} = C(12, 6) x^3 y^{-3}$

Sol.6. (c)

In the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$, let T_r is the n_{th} term

$T_r = {}^{15}C_{r-1} (x^4)^{15-r+1} \left(\frac{1}{x^3}\right)^{r-1}$

$= {}^{15}C_{r-1} x^{64-4-3r3+3} = {}^{15}C_{r-1} x^{67-7r}$

x^4 occurs in this term

$\Rightarrow 4 = 67 - 7r$

$\Rightarrow 7r = 63$

$\Rightarrow r = 9$

Sol.7. (c)

Given expression is:

$[(3x + y)^5]^4 - [3x - y)^4]^5 = [(3x + y)]^{20} - [(3x-y)]^{20}$

First and second expansion will have 21 terms each but odd terms in second expansion be 1^{st}, 3^{rd}, 5^{th} ...... 21^{st} will be equal and opposite to those of first expansion

Thus, the number of terms in the expansion o above expression is 10.

Sol.8. (c)

$T_r = {}^nC_{r-1} (2x)^{r-1} (3y^2)^{n-r+1}$

$T_4 = {}^5C_3 (2x)^3 (3y^2)^2$

$= \frac{5!}{3!2!} 2^3.x^3.9y^4 = \frac{5.4}{2.1} \times 8 \times 9 \times x^3 y^4 = 720\, x^3 y^4$

$\therefore$ coefficient $x^3y^4 = 720$

Sol.9. (d)

In 3^n, last digit is 3, if n = 1, 9 if n = 2, 7 if = 3 and1 if n = 4 and it is repeated after than

Given expression is $3^{3^{4n}} + 1$ d

Let $x = 3^{3^{4n}} + 1 = 3^{81n} + 1$

$\Rightarrow x = 3^{80n}.3^{n+1}$

Last digit of x will be decided by 3^n since 3^{80n} has power multiple of 4.

If n = 1 last digit is 3 + 1 = 4

n=2 last digit is $3^2+1=9+1=10$

So, last digit is zero

n=3 last digit is $3^3+1=27+1=28$

last digit is 8.

If n = 4 last digit is $3^4 + 1 = 81 + 1 = 82$

last digit is 2.

So, there is no definite value of last digit.

Sol.10. (d)

We find r_n term:

t_r is the r^{th} term in the expansion of $(1 + x)^{101}$.

$\therefore \frac{t_{20}}{t_{19}} = \frac{{}^{101}C_{19}}{{}^{101}C_{18}}.\frac{x^{19}}{x^{18}} = \frac{{}^{101}C_{19}x}{{}^{101}C_{18}}$

$= \frac{\frac{101!}{19!82!}}{\frac{101!}{18!83!}} x = \frac{83x}{19}$

Sol.11. (a)

$(1-x)^n = {}^nC_0 - {}^nC_1(x) + {}^nC_2 x^2 - {}^nC_3 x^3 + ... + (-1)^n\, {}^nC_n$

Put x=1 and n=8, then answer is 0

Sol.12. (b)

Given expansion

$(1+x+2x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$= (1+x+2x^3)\left[\left(\frac{3}{2}x^3\right)^9 - {}^9C_1\left(\frac{3}{2}x^2\right)^8 \right.$

$\left. + {}^9C_6\left(\frac{3}{2}x^2\right)^3\left(\frac{1}{3x}\right)^6 - {}^9C_7\left(\frac{3}{2}x^2\right)\left(\frac{1}{3x}\right)^7 ......\right]$

In the second bracket we have to search out terms of $x°$ and $\frac{1}{x^3}$ which when multiplied with the terms 1 and $2x^3$ in the first bracket will give a term independent of x. The term containing $\frac{1}{x}$ will not occur in the 2^{nd} bracket.

$\therefore$ Term independent of x.

$= 1 - \left[{}^9C_6 \frac{3^3}{2^3}.\frac{1}{3^6}\right] - 2x^3\left[{}^9C_7 \frac{3^2}{2^2}.\frac{1}{3^7}.\frac{1}{x^3}\right]$

$= \left[\frac{9.8.7}{1.2.3}.\frac{1}{8.27}\right] - 2\left[\frac{9.8}{1.2}.\frac{1}{4.243}\right]$

$= \frac{7}{18} - \frac{2}{27} = \frac{17}{54}$

Sol.13. (c)

Consider $(1 + 2x + 3x^2 + 4x^3 + ...)^{1/2} = (1 - x^{-2})^{1/2}$

As we know that $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + ..$

$\Rightarrow (1 - x)^{-1} = 1 + x + x^2 + x^3 + x^4 + ...$

$\therefore$ Required coefficient of x^4 is 1.

Sol.14. (a)

Statement I: Given expansion is $(1 + x)^8$
Since, n = 8 is even

$\therefore \left(\frac{n}{2}+1\right)$ th term is the middle term.

ie– $\left(\frac{8}{4}+1\right)^{th}$ = 5^{th} term = middle term

Now, 5^{th} term = ${}^8C_4 x^4 1^4 = {}^8C_4 x^4$

Coeff. of 5^{th} term (middle term) = ${}^8C_4$

Now, consider the expansion $\left(x+\frac{1}{x}\right)^8$

It's middle term = 5^{th} term

and 5^{th} term = ${}^8C_4 x^4 \left(\frac{1}{x}\right)^4 = {}^8C_4$

Hence, statement I is correct

Statement II : Coeff. Of middle term in $(x + 1)^8$ is ${}^8C_4 = \frac{8!}{4!4!} = 70$

Coeff of 5^{th} term in $(1+ x)^7$ = $7C_4 = \frac{7!}{4!3!} = 35$

Sol.15. **(c)**
No of zeroes

$= \left[\frac{80}{5}\right]+\left[\frac{80}{25}\right]+\left[\frac{80}{125}\right]$

=16 + 3 + 0 + …..
=19

Sol.16. (a)

$= \frac{16^{200}+5}{15} = \frac{(15+1)^{200}+5}{15}$

Remainder = 1+5=6

Sol.17. **(c)**
Sum of all coefficients = 2^n
Sum of odd place coefficients + sum of even place coefficient = 2^n and sum of odd place coefficients = sum of even place coefficients so sum of odd placed coefficients= $\frac{2^n}{2} = 2^{n-1}$

Sol.18. **(b)**
Let the term containing x^7 in the expansion of $\left(ax^2+\frac{1}{bx}\right)^8$ is T_{r+1}

$\therefore T_{r+1} = {}^8C_r (ax^2)^{8-r} \left(\frac{1}{bx}\right)^r = {}^8C_r \frac{a^{8-r}}{b^r} x^{163r}$

Since this term contains x^7.
$\therefore$ 16 –3r = 7
$\Rightarrow$ r = 3
$\therefore$Coefficient of x^7 in the expansion of

$\left(ax^2+\frac{1}{bx}\right) = {}^8C_3 .a^5 / b^3$

Sol.19. **(a)**
Also, the term containing x^{-7} in the expansion of $\left(ax-\frac{1}{bx^2}\right)^8$ is T_{R+1}.

$T_{R+1} = {}^8C_R (ax)^{8-R} \left(-\frac{1}{bx^2}\right)^R$

$=(-1)R\ {}^8C_R \frac{a^{8-R}}{b^R}.x^{8-3}R$

Since this term contains x^{-7}.
$\therefore$ 8 – 3R = –7 $\Rightarrow$ R = 5
$\therefore$ Coefficient of x^{-7} in the expansion of

$\left(ax-\frac{1}{bx^2}\right)^8 = (-1)^5\ {}^8C_5 \frac{a^3}{b^5}$

Sol.20. **(b)**

$\left(\frac{1-x}{1+x}\right)^2 = (1-x)^2(1+x)^{-2}$

$=(1–2x+x^2)(1–2x+3x^2–4x^3+5x^4 –…..)$
$=(5+8+3)x^4 + …..= 16x^4 + ……$

Sol.21. **(d)**
Required sum = $(45–49)^4 = (–4)^4 = 256$

Sol.22. **(d)**

$\frac{(n+1)(n+2)}{2}$

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NDA PYQ

1. What is the middle term in the expansion of $\left(1-\frac{x}{2}\right)^8$?

(a) $\frac{35x^4}{8}$ (b) $\frac{17x^5}{8}$
(c) $\frac{35x^5}{8}$ (d) None of these

[NDA (I) - 2011]

2. What is the ratio of coefficient of x^{15} to the term independent of x in $\left(x^2+\frac{2}{x}\right)^{15}$?

(a) 1/64 (b) 1/32
(c) 1/16 (d) 1/4

[NDA (II) - 2011]

3. For all $n \in N$, $2^{4n} - 15n - 1$ is divisible by:

(a) 125 (b) 225
(c) 450 (d) none of the above

[NDA (II) - 2011]

4. The value of the term independent of x in the expansion of $\left(x^2-\frac{1}{x}\right)^9$ is

(a) 9 (b) 18
(c) 48 (d) 84

[NDA (I) - 2012]

5. In the expansion of $(1 + x)^n$, what is the sum of even binomial coefficient?

(a) 2^n (b) 2^{n-1}
(c) 2^{n+1} (d) none of these

[NDA (I) - 2012]

6. What is the sum of the coefficients in the expansion of $(1 + x)^n$?

(a) 2^n (b) $2^n - 1$
(c) $2^n + 1$ (d) $n + 1$

[NDA (I) - 2013]

7. How many terms are there in the expansion of $(1 + 2x + x^2)^{10}$?

(a) 11 (b) 20
(c) 21 (d) 30

[NDA (II) - 2013]

8. If n be a positive integer and $(1 + x)^n = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$, then what is $a_0 + a_1 + a_2 + \ldots + a_n$ equal to?

(a) 1 (b) 2^n
(c) 2^{n-1} (d) 2^{n+1}

[NDA (II) - 2013]

Directions (for next three):

In the expansion of $\left(x^3-\frac{1}{x^2}\right)^n$, where n is a positive integer, the sum of the coefficients of x^5 and x^{10} is 0.

9. What is n equal to?

(a) 5 (b) 10
(c) 15 (d) None of these

[NDA (I) - 2014]

10. What is the value of the independent term?

(a) 5005 (b) 7200
(c) –5005 (d) –7200

[NDA (I) - 2014]

11. What is the sum of the coefficients of the two middle terms:

(a) 0 (b) 1
(c) –1 (d) None of these

[NDA (I) - 2014]

Directions (for next five):

Read the following information carefully and answer the questions given below.

Consider the expansion $\left(x^2+\frac{1}{x}\right)^{15}$.

12. What is the independent term in the given expansion?

(a) 2103 (b) 3003
(c) 4503 (d) None of these

[NDA (II)- 2014]

13. What is the ratio of coefficient of x^{15} to the term independent of x in the given expansion?

(a) 1 (b) 1/2
(c) 2/3 (d) 3/4

[NDA(II) - 2014]

14. What is the sum of the coefficients of the middle terms in the given expansion?

(a) C (15, 9) (b) C (16, 9)
(c) C (16, 8) (d) None of these

[NDA(II) - 2014]

15. Consider the following statements

I. There are 15 terms in the given expansion.
II. The coefficient of x^{12} is equal to that of x^3.

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA(II) - 2014]

16. Consider the following statements

I.The term containing x^2 does not exist in the given expansion.
II.The sum of the coefficients of all the terms in the given expansion is 2^{15}.

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) neither I nor II

[NDA(II) - 2014]

17. The coefficient of x^{99} in the expansion of $(x - 1)(x - 2)(x - 3) \ldots (x - 100)$ is

(a) 5050 (b) 5000
(c) –5050 (d) –5000

[NDA (II) - 2015]

18. In the expansion of $\left(\sqrt{x}+\frac{1}{3x^2}\right)^{10}$ the value of constant term (independent of x) is

(a) 5 (b) 8
(c) 45 (d) 90

[NDA (II) - 2015]

Directions (for next three):

Read the following information carefully and answer the questions given below.

Consider the expansion of $(1 + x)^{2n+1}$.

19. The sum of the coefficients of all the terms in the expansion is

(a) 2^{2n-1} (b) 4^{n-1}
(c) 2×4^n (d) None of these

[NDA (II) - 2015]

20. The average of the coefficients of the two middle terms in the expansion is

(a) ${}^{2n+1}C_{n+2}$ (b) ${}^{2n+1}C_n$
(c) ${}^{2n+1}C_{n-1}$ (d) ${}^{2n}C_{n+1}$

[NDA (II) - 2015]

21. If the coefficient of x^r and x^{r+1} are equal in the expansion, then r is equal to

(a) n (b) $\frac{2n-1}{2}$

(c) $\frac{2n+1}{2}$ (d) n + 1

[NDA (II) - 2015]

22. The expansion of $(x-y)^n$, $n \geq 5$ is done in the descending powers of x. If the sum of the fifth and sixth terms is zero, then $\frac{x}{y}$ is equal to:

(a) $\frac{n-5}{6}$ (b) $\frac{n-4}{5}$

(c) $\frac{5}{n-4}$ (d) $\frac{6}{n-5}$

[NDA (I) - 2017]

23. The number of terms in the expansion of $(x+a)^{100} + (x-a)^{100}$ after simplification is:
(a) 202 (b) 101
(c) 51 (d) 50

[NDA (II) - 2017]

24. In the expansion of $(1+x)^{50}$, the sum of the coefficient of odd powers of x is:
(a) 2^{26} (b) 2^{49}
(c) 2^{50} (d) 2^{51}

[NDA (II) - 2017]

25. If $n \in N$, then $121^n - 25^n + 1900^n - (-4)^n$ is divisible by which one of the following:
(a) 1904 (b) 2000
(c) 2002 (d) 2006

[NDA-2018(1)]

26. What is the number of non zero terms in the expansion of $(1+2\sqrt{3x})^{11} + (1-2\sqrt{3x})^{11}$?
(a) 4 (b) 5
(c) 6 (d) 11

[NDA (I) - 2018]

27. If the coefficient of a^m and a^n in the expansion of $(1+a)^{m+n}$ are α and β than which one of the following is correct?

(a) $\alpha = 2\beta$ (b) $\alpha = \beta$
(c) $2\alpha = \beta$ (d) $\alpha = (m+n)^\beta$

[NDA (I) - 2018]

28. In the expansion of $(1+x)^{43}$, if the coefficient of $(2r+1)^{th}$ and $(r+2)^{th}$ terms are equal, than what is the value of r ($r \neq 1$)?
(a) 5 (b) 14
(c) 21 (d) 22

[NDA (I) - 2018]

29. What is the greatest integer among the following by which the number $5^5 + 7^5$ is divisible?
(a) 6 (b) 8
(c) 11 (d) 12

[NDA (I) - 2018]

30. Let the coefficient of the middle term of the binomial expansion of $(1+x)^{2n}$ be α and those of two middle terms of the binomial expansion of $(1+x)^{2n-1}$ be β and r which one of the following relation is correct ?
(a) $\alpha > \beta + r$ (b) $\alpha < \beta + r$
(c) $\alpha = \beta + r$ (d) $\alpha = \beta r$

[NDA (II) - 2018]

31. What is the coefficient of the middle term in the binomial expansion of $(2+3x)^4$:
(a) 6 (b) 12
(c) 108 (d) 216

[NDA (II) - 2018]

32. Find the number of terms in the expansion of $[(2x - 3y)^2(2x + 3y)^2]^2$?
(a) 4 (b) 5
(c) 8 (d) 16

[NDA (I) - 2019]

33. In the expansion of $(1 + ax)^n$,first three terms are 1, 12x and $64x^2$, then n = ?
(a) 6 (b) 9
(c) 10 (d) 12

[NDA (I) - 2019]

34. How many terms are there in the expression of $(1 + 2x + x^2)^5 + (1 + 4y + 4y^2)^5$?
(a) 12 (b) 20
(c) 21 (d) 22

[NDA (II) - 2019]

35. If the middle term in the expression of $\left(x^2+\frac{1}{x}\right)^{2n}$ is $184756x^{10}$, then what is the value of n?
(a) 10 (b) 8
(c) 5 (d) 4

[NDA (II) - 2019]

36. If the constant term in the expansion of $\left(\sqrt{x}-\frac{k}{x^2}\right)^{10}$ is 405, then what can be the value of k?
(a) ±2 (b) ±3
(c) ±5 (d) ±9

[NDA (II) - 2019]

37. What is the sum of last five coefficients in the expansion of $(1+x)^9$ when it is expanded in ascending power of x?
(a) 256 (b) 512
(c) 1024 (d) 2048

[NDA 2020]

38. The term independent of x in the binomial expansion of $\left(\frac{2}{x^2}-\sqrt{x}\right)^{10}$ is equal to
(a) 180 (b) 120
(c) 90 (c) 72

[NDA 2020]

39. If $(1 + 2x - x^2)^6 = a_0 + a_1x + a_2x^2 + ... + a_{12}x^{12}$ then what is $a_0 - a_1 + a_2 - a_3 + ... + a_{12}$ equal to
(a) 32 (b) 64
(c) 2048 (d) 4096

[NDA 2020]

40. What is the coefficient of middle term in the expansion of $(1 + 4x + 4x^2)^5$?
(a) 8064 (b) 4032
(c) 2016 (d) 1008

[NDA (I) - 2021]

41. If C_0 , C_1 , C_2 , ...C_n are the coefficients in the expression of $(1 + x)^n$, then what is the value of $C_1 + C_2 + C_3 + ... + C_n$?
(a) 2^n (b) $2^n - 1$
(c) 2^{n-1} (d) $2^n - 2$

[NDA (I) - 2021]

42. What is the sum of the coefficients of first and last terms in the expansion of $(1 + x)^{2n}$, where n is a natural number ?
(a) 1 (b) 2

(c) n (d) 2n

[NDA (I) - 2021]

43. How many terms are there in the expansion of $\left(\frac{a^2}{b^2}+\frac{b^2}{a^2}+2\right)^{21}$ where $a \neq 0, b \neq 0$?

(a) 21 (b) 22
(c) 42 (d) 43

[NDA (II) - 2021]

44. Consider the expansion of $(1 + x)^n$. Let p, q, r and s be the coefficient of first, second, nth and $(n + 1)^{th}$ terms respectively. What is (ps + qr) equal to?

(a) 1 + 2n (b) $1 + 2n^2$
(c) $1 + n^2$ (d) 1 + 4n

[NDA (II) 2021]

45. How many terms are there in the expansion of $\left(1+\frac{2}{x}\right)^9 \times \left(1-\frac{2}{x}\right)^9$?

(a) 9 (b) 10
(c) 19 (d) 20

[NDA (I) - 2022]

46. Consider the following statement in respect of the expansion of $(x + y)^{10}$:

1.Among all the coefficients of the terms, the coefficient of the 6^{th} term has the highest value.

2.The coefficient of the 3^{rd} term is equal to coefficient of the 9th term.

Which of the above statement is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2022]

47. In the expansion of $\left(x+\frac{1}{x}\right)^{2n}$, what is the $(n + 1)^{th}$ term from the end (when arranged in descending powers of x)?

(a) C(2n, n)x (b) C(2n, n–1)x
(c) C(2n, n) (d) C(2n, n–1)

[NDA (I) - 2022]

48. What is $\sum_{r=o}^{n} 2^r C(n,r)$ equal to?

(a) 2^n (b) 3^n
(c) 2^{2n} (d) 3^{2n}

[NDA 2022 (II)]

Consider the following for the next three (03) items that follow:

Consider the binomial expansion of $(p+qx)^9$:

49. What is the value of q if the coefficients of x^3 and x^6 are equal?

(a) p (b) 9p
(c) $\frac{1}{p}$ (d) p^2

[NDA 2022 (II)]

50. What is the ratio of the coefficients of middle terms in the expansion (when expanded in ascending powers of x)?

(a) pq (b) p/q
(c) 4p/5q (d) 1/(pq)

[NDA 2022 (II)]

51. Under what condition of the coefficients of x^2 and x^4 are equal?

(a) p:q = 7:2 (b) $p^2:q^2 = 7:2$
(c) p:q = 2:7 (d) $p^2:q^2 = 2:7$

[NDA 2022 (II)]

52. How many terms are there in the expansion of $(3x–y)^4 (x+3y)^4$?

(a 9 (b)12
(c) 15 (d) 17

[NDA – 2023 (1)]

53. If the sixth term in the binomial expansion of $\left(x^{-\frac{8}{3}}+x^2 \log_{10} x\right)^8$ is 5600, then what is the value of x?

(a) 6 (b) 8
(c) 9 (d)10

[NDA – 2023 (1)]

Consider the following for the next (03) items that follow:

Let $(1+x)^n = 1 + T_1x + T_2x^2 + T_3x^3 + \ldots\ldots + T_nx^n$.

54. What is $T_1 + 2T_2 + 3T_3 + \ldots + nT_n$ equal to?

(a) 0 (b) 1
(c) 2^n (d) $n2^{n-1}$

[NDA-2023 (2)]

55. What is $1–T_1 + 2T_2 – 3T_3 + \ldots + (–1)^n nT_n$ equal to?

(a) 0 (b) $–2^{n-1}$
(c) $n2^{n-1}$ (d) 1

[NDA-2023 (2)]

56. What is $T_1 + T_2 + T_3 + \ldots + T_n$ equal to?

(a) 2^n (b) $2^n – 1$
(c) 2^{n-1} (d) $2^n + 1$

[NDA-2023 (2)]

57. What is the coefficient of x^{10} in the expansion of $\left(1-x^2\right)^{20}\left(2-x^2-\frac{1}{x^2}\right)^{-5}$?

(a) –1
(b) 1
(c) 10
(d) Coefficient of x^{10} does not exist

[NDA-2024 (1)]

58. What is the remainder when 2^{120} is divided by 7?

(a) 1 (b) 3
(c) 5 (d) 6

[NDA-2024 (1)]

59. If the 4^{th} term in the expansion of $\left(mx+\frac{1}{x}\right)^n$ is $\frac{5}{2}$, then what is the value of mn?

(a) –3 (b) 3
(c) 6 (d) 12

[NDA-2024 (1)]

60. In a binomial expansion of $(x + y)^{2n+1} (x – y)^{2n+1}$, the sum of middle terms is zero. What is the value of $\left(\frac{x^2}{y^2}\right)$?

(a) 1 (b) 2
(c) 4 (d) 8

[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below

Let $(8 + 3\sqrt{7})^{20} = U + V$ and $(8 – 3\sqrt{7})^{20} = W$, where U is an integer and 0 < V < 1.

61. What is V + W equal to ?

(a) 8 (b) 4
(c) 2 (d) 1

[NDA-2024 (2)]

62. What is the value of (U + V)W ?

(a) 1/2 (b) 1
(c) 3/2 (d) 2

[NDA-2024 (2)]

63. What is the remainder when $7^n - 6n$ is divided by 36 for n = 100?

(a) 0 (b) 1
(c) 2 (d) 6

[NDA-2024 (2)]

64. In the expansion of $(1 + x)^p(1 + x)^q$, if the coefficient of x^3 is 35, then what is the value of (p + q)?

(a) 5 (b) 6
(c) 7 (d) 8

[NDA-2024 (2)]

65. What is the number of rational terms in the expansion of $\left(3^{\frac{1}{2}}+5^{\frac{1}{4}}\right)^{12}$?

(a) 2 (b) 4
(c) 6 (d) 12

[NDA-2025 (1)]

66. If the sum of binomial coefficients in the expansion of $(x + y)^n$ is 256, then the greatest binomial coefficient occurs in which one of the following terms?

(a) Third (b) Forth
(c) Fifth (d) Ninth

[NDA-2025 (1)]

67. If $k<\left(\sqrt{2}+1\right)^3<k+2$, where k is a natural number, then what is the value of k?

(a) 11 (b) 13
(c) 15 (d) 17

[NDA-2025 (1)]

68. What is the remainder when 5^{99} is divided by 13?

(a) 10 (b) 9
(c) 8 (d) 6

[NDA-2025 (2)]

69. Which one of the following is the greatest coefficient in the expansion of $((1 + x)^{100}$?

(a) The coefficient of x^{100} (b) The coefficient of x^{99}
(c) The coefficient of x^{51} (d) The coefficient of x^{50}

[NDA-2025 (2)]

ANSWER KEY

1.	a	**2.**	b	**3.**	b	**4.**	d	**5.**	b	**6.**	a	**7.**	c	**8.**	b	**9.**	c	**10.**	c
11.	a	**12.**	b	**13.**	a	**14.**	c	**15.**	b	**16.**	c	**17.**	c	**18.**	a	**19.**	c	**20.**	b
21.	a	**22**	b	**23.**	c	**24.**	b	**25.**	b	**26.**	c	**27.**	b	**28.**	b	**29.**	d	**30.**	c
31.	d	**32.**	b	**33.**	b	**34.**	c	**35.**	a	**36.**	b	**37.**	a	**38.**	a	**39.**	b	**40.**	a
41.	b	**42.**	b	**43.**	d	**44.**	c	**45.**	b	**46.**	c	**47.**	c	**48.**	b	**49.**	a	**50.**	b
51.	b	**52.**	a	**53.**	d	**54.**	d	**55.**	d	**56.**	b	**57.**	a	**58.**	a	**59.**	b	**60.**	a
61.	d	**62.**	b	**63.**	b	**64.**	c	**65.**	b	**66.**	c	**67.**	b	**68.**	c	**69.**	d		

Solutions

Sol.1. (a)

Given expression = $\left(1-\frac{x}{2}\right)^8$

Here, n = 8, which is an even number

So, $\left(\frac{8}{2}+1\right)$ the term is the middle term.

Now, $T_5 = {}^8C_4(1)^{8-4}\left(-\frac{x}{2}\right)^4$

$=\frac{8.7.6.5}{4.3.2.1}\left(\frac{x^4}{16}\right)=\frac{35x^2}{8}$

Sol.2. (b)

Binomial expression = $\left(x^2+\frac{2}{x}\right)^{15}$

General term,

$T_{r+1} = {}^{15}C_r\,(x^2)^{15-r}\left(\frac{2}{x}\right)^r$

$= {}^{15}C_r\,x^{30-2r}\,(2)^r\,.\,x^{-r} = {}^{15}C_r\,2^r\,x^{30-3r}$ …(i)

Put $30 - 3r = 15 \Rightarrow 3r = 15 \Rightarrow r = 5$

$\therefore t_{5+1} = {}^{15}C_5.\,2^5.\,X^{30-30\times5} = {}^{15}C_5\,2^5\,.\,x^{15}$

Coefficient of x^{15} in $\left(x^2+\frac{2}{x}\right)^{15}={}^{15}C_r.2^5$ …(ii)

For coefficient of c^0,

Put, $30-3r = 0 \Rightarrow r = 10$

$\therefore T_{10+1} = {}^{15}C_{10}\,2^{10}.\,X^{30-3\times10} = {}^{15}C_{10}\,2^{10}.\,x^0$…(iii)

$\therefore$Ratio of coefficient of x^{15} to the term independent of x

$=\frac{{}^{15}C_5.2^5}{{}^{15}C_{10}2^{10}}=\frac{{}^{15}C_{10}.2^5}{{}^{15}C_{10}.2^{10}}=\frac{1}{2^5}=\frac{1}{32}$

$(\because {}^nC_r = {}^nC_{n-r})$

Sol.3. (b)

$2^{4n} - 15n - 1 = 16^n - 15n - 1$

$= (15 + 1)^n - 15n - 1$

$=[{}^nC_0 15^n + {}^nC_1 15^{n-1} + \ldots + {}^nC_{n-2} 15^2 + {}^nC_1 15 + {}^nC_n] - 15n - 1$

$=[{}^nC_0 15^n + {}^nC_1 15^{n-1} + \ldots + {}^nC_{n-2} 15^2]$

15^2 common so 225 will always divide the expression.

Sol.4. (d)

Given, $\left(x^2-\frac{1}{x}\right)^9$

General term, $T_{r+1} = {}^nC_r\,(1)^{n-r}\,x^r$, in $(1+x)^n$

Similarly,

$T_{r+1} = {}^9C_r\,(x^2)^{9-r}.\left(-\frac{1}{x}\right)^r$, in $\left(x^2-\frac{1}{x}\right)^9$

$= {}^9C_r\,x^{18-2r}\,(-1)^r\,x^{-r}$

$= {}^9C_r\,x^{(18-3r)}\,x^{-r}$ …(i)

For independent term,

Put $18 - 3r = 0 \Rightarrow 3r = 18 \Rightarrow r = 6$

$\therefore T_{(6+1)} = {}^9C_6\,x^{(18-18)}.\,(-1)^6$

$T_7 = {}^9C_6\,.\,1=\frac{9.8.7}{3.2.1}=84$

Sol.5. (b)

sum of even or odd coefficient = 2^{n-1}

Sol.6. (a)

For the sum of the coefficients in the expansion of $(1+x)^n$

Put x =1

$(1+x)^n = (1+1)^n = 2^n$

Which is the required sum of the coefficient.

Sol.7. (c)

Given, $(1+2x+x^2)^{10} = \{(1+x)^2\}^{10} = (1+x)^{20}$

$\therefore$ total terms = 20 + 1 = 21

Sol.8. (b)

Given that,

$(1+x)^n = a_0 + a_1\,x + a_2\,x^2 + \ldots + a_n\,x^n$

Put x = 1

$(1+1)^n = a_0 + a_1 + a_2 + \ldots + a_n$

$\Rightarrow a_0 + a_1 + a_2 + \ldots\ldots + a_n = 2^n$

Sol.9. (c)

$${}^nC_{\left(\frac{3n}{2}-2\right)} = {}^nC_{\left(\frac{3n}{5}-1\right)} \quad \text{[from Equation (ii)]}$$

$$\Rightarrow n\left(\frac{3n}{5}-2\right)+\left(\frac{3n}{5}-1\right)$$

$(\because {}^nC_x = {}^nC_y \Rightarrow n = x + y)$

$$\Rightarrow n = \frac{6n}{5} - 3$$

$$\Rightarrow \frac{6n}{5} - n = 3 \Rightarrow \frac{n}{5} = 3$$

$\therefore$ n = 15

Sol.10. (c)

For the independent term,

Put $3n - 5r = 0$ [from eq. (i)]

$\Rightarrow 5r = 3n = 3 \times 15$ $(\because n = 15)$

$\Rightarrow 5r = 3 \times 3 \times 5$

$\therefore r = 9$

Now, put the value of r in Eq. (i), we get,

$T_{9+1} = {}^{15}C_9\,(-1)^9 .\ x^{(3\times15-5\times9)}$

$\Rightarrow T_{10} = -\,{}^{15}C_9 .\ x^0 = -\,{}^{15}C_9$

$\Rightarrow T_{10} = -\,{}^{15}C_6$ $(\because {}^nC_r = {}^nC_{n-r})$

$$= -\frac{15!}{6!9!}\left(\because \quad \frac{n!}{r!(n-r)!}\right)$$

$$= -\frac{15\times14\times13\times12\times11\times10}{6\times5\times4\times3\times2\times1}$$

$= -5005$

Sol.11. (a)

Since, n = 15

$\therefore$ Total term in the expansion of $\left(x^3 - \frac{1}{x^2}\right)^{15}$ is 16

So, middle term $= \left(\frac{n}{2}\right)$ th term and $\left(\frac{n}{2}+1\right)$ th term

$= \left(\frac{16}{2}\right)$ th term and $\left(\frac{16}{2}+1\right)$ th term

$= 8^{th}$ term and 9^{th} term

So, $\left(x^3 - \frac{1}{x^2}\right)^{15}$ has two middle terms

$T_8 = T_{(7+1)} = {}^{15}C_8 .\ (-1)^7 .\ x^{(3\times15-5\times7)}$

$= -\,{}^{15}C_7 .\ x^{10}$ [from eq. (i)]

and $T_9 = T_{(8+1)}\ {}^{15}C_8 .\ (-1)^8 .\ x^{(3\times15-5\times8)}$

$= {}^{15}C_8 .\ x^5$ [from eq. (ii)]

Now, the sum of the coefficients of the two middle terms

$= -\,{}^{15}C_7 + {}^{15}C_8$

$= -\,{}^{15}C_7 + {}^{15}C_7$ $(\because {}^nC_r = {}^nC_{n-r})$

$= 0$

Sol.12. (b)

We have, $\left(x^2 + \frac{1}{x}\right)^{15}$

$$T_{r+1} = {}^{15}C_r\,(x^2)^{15-r}\left(\frac{1}{x}\right)^r$$

$= {}^{15}C_r\, x^{30-2r-r} = {}^{15}C_r\, x^{30-3r}$

For independent term,

$30 - 3r = 0 \Rightarrow r = 100$

Put r = 10, we get

$$T_{10+1} = {}^{15}C_{10} = \frac{15!}{10!5!}$$

$$= \frac{15\times14\times13\times12\times11\times10!}{10!\times1\times2\times3\times4\times5} = 3003$$

Sol.13. (a)

For coefficient of x^{15}

$30 - 3r = 15 \Rightarrow r = 5$

So, the coefficient of x^{15} is ${}^{15}C_5$.

and coefficient of independent of x is

$30 - 3r = 0 \Rightarrow r = 10$

So, coefficient of independent of x is ${}^{15}C_{10}$.

$$\therefore \text{Required ratio} = \frac{{}^{15}C_5}{{}^{15}C_{10}}$$

$$= \frac{{}^{15}C_5}{{}^{15}C_5} \quad (\because {}^nC_r = {}^nC_{n-r}) = 1$$

Sol.14. (c)

We have, $\left(x^2 + \frac{1}{x}\right)^{15}$

Since, n is odd

So, it has two middle terms T_8 and T_9

$\therefore\ T_8 + T_9 = {}^{15}C_7 + {}^{15}C_8$

$= {}^{16}C_8$ $(\because {}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r)$

Sol.15. (b)

1.We know that, $(a+b)^n$ have total number of terms is n + 1.

So, $\left(x^2 + \frac{1}{x}\right)^{15}$ have 16 terms

Hence, Statement 1 is not correct.

2.For coefficient of x^{12},

$30 - 3r = 12$

$\Rightarrow r = 6 \Rightarrow {}^{15}C_6$

and for coefficient of x^3,

$30 - 3r = 3 \Rightarrow r = 9 \Rightarrow {}^{15}C_9$

$\therefore {}^{15}C_6 = {}^{15}C_9$

Hence, statement 2 is correct.

Sol.16. (c)

1. For coefficient of x^2

$30 - 3r = 2 \Rightarrow r = \frac{28}{3}$, $r \notin N$

So, x^2 does not exist in the expansion.

Hence, statement 1 is correct

2. Now,

$$\left(x^2 + \frac{1}{4}\right)^{15} = {}^{15}C_0\left(x^2\right)^{15} + {}^{15}C_1\left(x^2\right)^{14}\left(\frac{1}{x}\right) + \ldots + {}^{15}C_{15}\left(\frac{1}{x}\right)^{15}$$

Put x = 1 both sides, we get

$(1+1)^{15} = {}^{15}C_0 + {}^{15}C_1 + \ldots + {}^{15}C_{15}$

$\Rightarrow 2^{15} = {}^{15}C_0 + {}^{15}C_1 + \ldots + {}^{15}C_{15}$

Hence, statement 2 is correct.

Sol.17. (c)

$(x-1)(x-2)(x-3)\ldots(x-100)$

Then the coefficient of x^{99} is given as

$-1-2-3-4 \ldots\ldots -100$

$= -1\,(1+2+3+\ldots 100)$

$$= -\left(100\times\frac{101}{2}\right)$$

$= (-5050)$

Sol.18. (a)

Independent term in the expansion of $\left(\sqrt{x} + \frac{1}{3x^2}\right)^{10}$

$$T_{r+1} = {}^{10}C_r\, x^{1/2(10-r)} \times \left(\frac{1}{3x^2}\right)^r$$

$$= {}^{10}C_r .\ x^{1/2(10-r)-2r}\left(\frac{1}{3}\right)^r \quad \ldots(i)$$

Term independent from x

$$= \frac{1}{2}(10-r) - 2r = 0$$

$\Rightarrow 10 - r - 4r = 0$

$\Rightarrow r = 2$

Putting the value of r in eq. (ii)

$$T_{2+1} = {}^{10}C_2 \times \left(\frac{1}{3}\right)^2 = \frac{10\times9\times8!}{2\times8!}\times\frac{1}{9} = 5$$

Sol.19. (c)

The sum of the coefficient $= 2^{2n+1} = (2^2)^n \times 2 = 2 \times 4^n$.

Sol.20. (b)

Here total number of terms = 2n + 1 + 1 = 2n + 2

$\therefore$ middle terms are $\left(\frac{2n+2}{2}\right)$ th term and $\left(\frac{2n+4}{2}\right)$ th term

i.e., (n+1) th and (n+2) th term

Now, $T_{n+1} = {}^{2n+1}C_n .\ x^n$

and $T_{n+2} = {}^{2n+1}Cn_{+1} .\ x^{n+1}$

$$\therefore \text{required average} = \frac{{}^{2n+1}C_n + {}^{2n+1}C_{n+1}}{2}$$

$$= \frac{{}^{2n+1}C_n + {}^{2n+1}C_{n+1}}{2}$$

{$\because$ these two coefficients are equal.}

$= {}^{2n+1}C_n$.

Sol.21. (a)

$(1+x)^{2n+1}$

Given the coefficient of x^r and x^{r+1} are equal

$T_{r+1} = {}^{2n+1}C_r .(1)^{2n+1-r} .\ (x)^r$

$= {}^{2n+1\,1}C_r .(1).\ x^r$ …(i)

The coefficient of $x^r = {}^{2x+1}C_r$

and the coefficient of x^{r+1}

$= {}^{2n+1}C_{r+1} .(1)^{2n+1-r-1}$

$= {}^{2n+1}C_{r+1}$ …(ii)

According to the question,

the coefficient of x^r and x^{r+1} are equal

$\Rightarrow {}^{2n+1}C_r = {}^{2n+1}C_{r+1}$

$$\Rightarrow \frac{(2n+1)!}{2n+1-r!} = \frac{(2n+1)!}{(2n-r)!(r+1)!}$$

$\Rightarrow (2n-r)!(r+1)! = (2n+1-r)!(r)!$

$\Rightarrow (2n-r)!(r+1)r! = (2n+1-r)\,(2n-r)!r!$

$\Rightarrow r + 1 = 2n + 1 - r$

$\Rightarrow 2r = 2n$

$\Rightarrow r = n$

Sol.22. (b)

In the expansion of $(x-y)^n$, in decreasing power of x,

$T_5 = T_{4+1} = {}^nC_4 .\ x^{n-4} .\ (-y)^5$

and $T_6 = T_{5+1} = {}^nC_5 .\ x^{n-5} .\ (-y)^5$

According to the question,

$T_5 + T_6 = 0$

$\Rightarrow {}^nC_4 .n^{n-4} .y^4 + {}^nC_5 .\ x^{n-5} .\ (-y)^5 = 0$

$\Rightarrow {}^nC_4 .n^{n-4} .y^4 = {}^nC_5 .\ x^{n-5} .\ y^5$

$$\Rightarrow \frac{x^{n-4}}{x^{n-5}} . \frac{y^4}{y^5} = \frac{{}^nC_5}{{}^nC_4}$$

$$\Rightarrow \frac{x}{y} = \frac{n!}{(n-5)!,5!} \times \frac{(n-4)!4!}{n!} = \frac{n-4}{5}$$

Sol.23. (c)

The number of terms in the expansion of $(x+a)^n + (x-a)^n$. If n is even equal to $\left(\frac{n}{2}+1\right)$ i.e., $\left(\frac{100}{2}+1\right) = 51$

Sol.24. (b)
In $(1+x)^{50}$ the coefficient of odd powers $= \frac{2^{50}}{2} = 2^{49}$
Sol.25. (b)
$121^n - 25^n + 1900^n - (-4)^n$
Put n = 1 , we will get 2000
Which is divisible by 2000 according to second option
Sol.26. (c)
$(a+b)^{11} + (a-b)^{11}$
$\Rightarrow$(6 even terms + 6 even term) + (6 even terms – 6 odd terms)
$\Rightarrow$ 6 terms
Sol.27. (b)
$(1+a)^m (1+a)^n$
$({}^mC_0 + {}^mG_1\, a + {}^mC_2 a^2 + \ldots {}^mC_m\, a^m)\ ({}^nC_0 + {}^nC_1\, a^2 + \ldots\ldots {}^nC_n\, a^n)$
Coefficient of $a^m = {}^mC_m \cdot {}^nC_0 = 1 \Rightarrow \alpha$
Coefficient of $a^n = {}^nC_n \cdot {}^mC_0 = 1 \Rightarrow \beta$
Sol.28. (b)
$(2r+1) + (r+2) = 43 + 2,\ 3r + 3 = 45$
$r = \frac{42}{3} = 14$
Sol.29. (d)
$(a^n + b^n)$ is always divisible by a + b when n is odd.
Sol.30. (c)
Coefficient of middle term of $(1+x)^{2n} = {}^{2n}C_n = \alpha$
Coefficient of middle term of $(1+x)^{2n-1}$
$= {}^{2n-1}C_{n-1} = \beta$ and ${}^{2n-1}C_n = r$
$\because {}^{2n-1}C_{n-1} + {}^{2n-1}C_n = {}^{2n}C_n$
$\beta + r = \alpha$
Sol.31. (d)
Middle term is 3rd term
$T_3 = {}^4C_2\,(2)^2\,(3x)^2 = \frac{4\times 3}{2} \times 4 \times 9x^2$
$= 216x^2$
Coefficient =216
Sol.32. (b)
$[(2x-3y)^2(2x+3y)^2]^2 = (4x^2 - 9y^2)^4$
no. of terms in expansion = 5
Sol.33. (b)
In the expansion of $(1+ax)^n$,first three terms are 1, 12x and $64x^2$, then n = ?
$(1+ax)^n = 1 + nax + \frac{n(n-1)}{2}a^2x^2 + \ldots..$
$\Rightarrow$ an = 12 and $\frac{n(n-1)}{2}a^2 = 64$
by solving these equations n = 9
Sol.34. (c)
$(1+x)^{10} + (1+2y)^{10}$
11 terms in each expression but 1 term constant is common so there is 21 terms total.
Sol.35. (a)
Middle term is (n + 1)th term
$T_{n+1} = {}^{2n}C_n\,(x^2)^n\,(1/x)^n$
$= {}^{2n}C_n (x)^n = 184756x^{10}$
n = 10 by comparing the powers of x.
Sol.36. (b)
General term, $T_{r+1} = {}^nC_r\,(a)^{n-r}\,(b)^r$, in $(a+b)^n$
$T_{r+1} = {}^{10}C_r\,(\sqrt{x})^{10-r}.\left(-\frac{k}{x^2}\right)^r,$
$= {}^{10}C_r\, x^{5-r/2}\,(-k)^r\, x^{-2r}$ …(i)
For independent term,
Put $5 - r/2 - 2r = 0 \Rightarrow r = 3$

$T_{2+1} = {}^{10}C_2\,(\sqrt{x})^{10-2}.\left(-\frac{k}{x^2}\right)^2,$
$45k^2 = 405$
$k = \pm 3$
Sol.37. (a)
sum of coefficients of all terms = $2^9 = 512$
there is total 10 terms and sum of first five terms = sum of last five terms
so sum of last five terms = 256
Sol.38. (a)
$\left(\frac{2}{x^2} - \sqrt{x}\right)^{10}$
General term, $T_{r+1} = {}^nC_r\,(a)^{n-r}\,(b)^r$, in $(a+b)^n$
$T_{r+1} = {}^{10}C_r\left(\frac{2}{x^2}\right)^{10-r}\cdot\left(\sqrt{x}\right)^r$
$= {}^{10}C_r\,.\,2^{10-r}\ x^{20-2r}\,(x)^{r/2}$ …(i)
For independent term,
Put $20 - 5r/2 = 0 \Rightarrow r = 8$
$T_{8+1} = {}^{10}C_8\left(\frac{2}{x^2}\right)^{10-8}\cdot\left(\sqrt{x}\right)^8 = 45.2^2$
= 180
Sol.39. (b)
If $(1 + 2x - x^2)^6 = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{12}x^{12}$
put x = −1 both sides
$a_0 - a_1 + a_2 - a_3 + \ldots + a_{12} = (1-2-1)^6 = 64$
Sol.40. (a)
$(1+4x+4x^2)^5 = \mathbf{(1+2x)^{10}}$
There will be 11 terms in expansion, and 6th term will be its middle term
$\Rightarrow {}^{10}C_5(1)^5.(2x)^5 = 8064x^2$
Coefficient of middle term is 8064
Sol.41. (b)
${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots + {}^nC_n = 2^n$
${}^nC_1 + {}^nC_2 + \ldots + {}^nC_n = 2^n{}_{-1}$
Sol.42. (b)
$(1+x)^{2n}$
coeff. of first and last terms is 1 and 1
so sum is 2
Sol.43. (d)
$\left(\frac{a^2}{b^2} + \frac{b^2}{a^2} + 2\right)^{21} = \left[\left(\frac{a}{b} + \frac{b}{a}\right)^2\right]^{21} = \left(\frac{a}{b} + \frac{b}{a}\right)^{42}$
Numbers of terms = 43
Sol.44. (c)
$(1+x)^n$
1st term = 1
2nd term = ${}^nC_1.x. = nx$
nth term = ${}^nC_{n-1}\, x^{n-1} = nx^{n-1}$
(n+1)th term = ${}^nC_n\, x^n = x^n$
p=1, q=n, r=n, S=1
PS + qr = $1 + n^2$
Sol.45. (b)
$\left(1 + \frac{2}{n}\right)^9\left(1 - \frac{2}{x}\right)^9 = \left(1 - \frac{4}{x^2}\right)^9$
total 10 terms in expansion.
Sol.46. (c)
1.middle term of $(x+y)^{10}$ is 6th term and we know that coefficient of middle term is largest.
2.coeficient of 3rd term = ${}^{10}C_2$ coefficient of 9th term – ${}^{10}C_8\ {}^{10}C_2 = {}^{10}C_8$
Both statements are correct.
Sol.47. (c)
(n + 1)th term of $\left(n + \frac{1}{x}\right)^{2n}$
$= {}^{2n}C_n\, x^n\,(\frac{1}{x})^n$
$= {}^{2n}C_n$

Sol.48. (b)
$\sum_{r=o}^{n} 2^r C(n,r) 2^n$
$= {}^nC_0 2^0. + . {}^nC_1.2^1 + {}^nC_2.2^2. + \ldots\ldots {}^nC_n$
$(a+b)^n = {}^nC_0 a^n. + . {}^nC_1.a^{n-1}b + {}^nC_2.a^{n-2}b^2.$
$+ \ldots\ldots\ldots\ldots + {}^nC_n b^n$
Put a = 1 b = 2
$\Rightarrow (1+2)^n = {}^nC_0 2^0. + . {}^nC_1.2^1 + {}^nC_2.2^2. + \ldots\ldots {}^nC_n 2^n$
$\Rightarrow 3^n = {}^nC_0 2^0. + . {}^nC_1.2^1 + {}^nC_2.2^2. + \ldots\ldots {}^nC_n 2^n$
Sol.49. (a)
General term (r+1)th term is
$\Rightarrow {}^nC_r p^{n-r}.q^r x^r$
For Coefficient of x^3 , r = 3
Then coeff of x^3 is $\Rightarrow {}^9C_3 p^{9-3}.q^3$
For Coefficient of x^6 , r = 6
Then coeff of x^6 is $\Rightarrow {}^9C_6 p^{9-6}.q^6$
coefficients of x^3 and x^6 are equal
$\Rightarrow {}^9C_3.p^6.q^3 = \Rightarrow {}^9C_6.p^3.q^6$
$\Rightarrow p^3 = q^3$
$\Rightarrow p = q$
Sol.50. (b)
Middle term exit at 5th and 6th position
5th term
$\Rightarrow {}^9C_4 p^{9-4}.q^4 = {}^9C_4 p^5.q^4$
6th term
$\Rightarrow {}^9C_5 p^{9-5}.q^5 = {}^9C_5 p^4.q^5$
Ratio of above two expressions is
$\Rightarrow \frac{{}^9C_4 p^5.q^4}{{}^9C_5 p^4.q^5} = \frac{p}{q}$
Sol.51. (b)
General term (r+1)th term is $\Rightarrow {}^nC_r p^{n-r}.q^r x^r$
For Coefficient of x^2 , r = 2
Then coeff of x^2 is $\Rightarrow {}^9C_2 p^{9-2}.q^2 = {}^9C_2 p^7.q^2$
For Coefficient of x^4 , r = 4
Then coeff of x^4 is $\Rightarrow {}^9C_4 p^{9-4}.q^4 = {}^9C_4 p^5.q^4$
coefficients of x^2 and x^4 are equal
$\Rightarrow {}^9C_2 p^7.q^2 = {}^9C_4 p^5.q^4$
$\Rightarrow 36p^2 = 126q^2$
$\Rightarrow 2p^2 = 7q^2$
$\Rightarrow p^2 : q^2 = 7 : 2$
Consider the binomial expansion of $(p+qx)^9$
Sol.52. (a)
Maximum power of x in the expansion of $(3x-y)^4\,(x+3y)^4$ is x^8 and minimum power of x will be x^0
When we write all these terms in descending power of x , we will get 9 terms.
Sol.53. (d)
Given
$\left(x^{-\frac{8}{3}} + x^2 \log_{10} x\right)^8$
6th term of above binomial expansion $(a+b)^n$ is

$\Rightarrow {}^8C_5\left(x^{-\frac{8}{3}}\right)^3 \cdot \left(x^2 \log_{10} x\right)^5$

$\Rightarrow 56\left(x^{-8}\right)x^{10}\left(\log_{10} x\right)^5 = 5600$

$\Rightarrow x^2\left(\log_{10} x\right)^5 = 100$

Now by checking options we can say x = 10

Sol.54. (d)

$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1}b + {}^nC_2 a^{n-2}b^2 + \ldots\ldots\ldots + {}^nC_n b^n$

$(a+b)^n = T_0 a^n + T_1 a^{n-1}b + T_2 a^{n-2}b^2 \ldots + \ldots\ldots\ldots + T_n b^n$

$(1+x)^n = T_0 + T_1 x + T_2 x^2 + T_3 x^3 \ldots + T_n x^n$

Differentiate both side w.r.t. x

$n(1+x)^{n-1} = T_1 + 2T_2 x + 3T_3 x^2 \ldots + nT_n x^{n-1}$ ……….(i)

Put x = 1

$n(1+1)^{n-1} = T_1 + 2T_2 + 3T_3 \ldots + nT_n$

$n2^{n-1} = T_1 + 2T_2 + 3T_3 \ldots + nT_n$

Sol.55. (d)

From equation (i) in above question

$n(1+x)^{n-1} = T_1 + 2T_2 x + 3T_3 x^2 \ldots + nT_n x^{n-1}$

Put x = -1

$0 = T_1 - 2T_2 + 3T_3 - \ldots + (-1)^n nT_n$

Multiply both side with – 1

$0 = -T_1 + 2T_2 - 3T_3 + \ldots + (-1)^n nT_n$

Add 1 both side

$1 = 1 - T_1 + 2T_2 - 3T_3 + \ldots + (-1)^n nT_n$

Sol.56. (b)

$(1+x)^n = T_0 + T_1 x + T_2 x^2 + T_3 x^3 \ldots + T_n x^n$

put x = 1

$2^n = T_0 + T_1 + T_2 + T_3 \ldots + T_n$

$2^n - 1 = T_1 + T_2 + T_3 \ldots + T_n$

Sol.57. (a)

$(1-x^2)^{20}\left(2 - x^2 - \frac{1}{x^2}\right)^{-5}$

$(1-x^2)^{20}\left(-\left(x - \frac{1}{x}\right)^2\right)^{-5}$

$-\frac{(1-x^2)^{20}}{\left(x - \frac{1}{x}\right)^{10}} = -\frac{(x^2-1)^{20}}{\left(\frac{x^2-1}{x}\right)^{10}}$

$= -(x^2-1)^{10} x^{10}$

In expansion of $(x^2 - 1)^{10}$ constant term is last term that is 1

So coefficient of x^{10} is – 1.

Sol.58. (a)

$\frac{2^{120}}{7} = \frac{8^{40}}{7} = \frac{(7+1)^{40}}{7}$

Constant term is remainder that is 1.

Sol.59. (b)

$\left(mx + \frac{1}{x}\right)^n$

4^{th} term

${}^nC_3 (mx)^{n-3} \cdot \left(\frac{1}{x}\right)^3 = \frac{5}{2}$

${}^nC_3 (m)^{n-3} \cdot (x)^{n-6} = \frac{5}{2}$

Given term is constant term so n – 6 = 0

So n = 6

${}^6C_3 (m)^3 = \frac{5}{2}$

$(m)^3 = \frac{1}{8} \Rightarrow m = \frac{1}{2}$

So mn = 3

Sol.60. (a)

$(x+y)^{2n+1}(x-y)^{2n+1}$

$(x^2 - y^2)^{2n+1}$

Power is odd so there will be 2 middle terms

1^{st} at $(n+1)^{th}$ position and next at $(n+2)^{th}$ position

Sum of middle terms is given zero

${}^{2n+1}C_n (x^2)^{2n+1-n}(-y^2)^n + {}^{2n+1}C_{n+1}(x^2)^{2n+1-n-1}(-y^2)^{n+1} = 0$

$\Rightarrow {}^{2n+1}C_n (x^2)^{n+1}(-y^2)^n + {}^{2n+1}C_{n+1}(x^2)^n(-y^2)^{n+1} = 0$

$\Rightarrow (x^2)^{n+1}(-y^2)^n + (x^2)^n(-y^2)^{n+1} = 0$

$\Rightarrow (x^2)^{n+1}(y^2)^n(-1)^n = -(x^2)^n(y^2)^{n+1}(-1)^{n+1}$

$\Rightarrow (x^2)^n x^2 (y^2)^n = (x^2)^n (y^2)^n y^2$

$\Rightarrow x^2 = y^2$

$\Rightarrow \frac{x^2}{y^2} = 1$

Sol.61. (d)

$(8 + 3\sqrt{7})^{20} = U + V$

$(8 - 3\sqrt{7})^{20} = W$

by adding both equations

U + V + W

$= (8 + 3\sqrt{7})^{20} + (8 - 3\sqrt{7})^{20} =$ Integer

All irrational terms are cancelled out.

U is given in question so V + W must be integer.

Now by rationalize W

$W = \left(8 - 3\sqrt{7} \times \frac{8+3\sqrt{7}}{8+3\sqrt{7}}\right)^{20} = \left(\frac{1}{8+3\sqrt{7}}\right)^{20}$

by above equation 0 < W < 1

and given that 0 < V < 1

by adding above two inequalities

0 < V + W < 2

so V + W = 1 (because V + W = Integer)

Sol.62. (b)

$(8 + 3\sqrt{7})^{20} = U + V$

$(8 - 3\sqrt{7})^{20} = W$

multiply both equations

$(U + V)W = (8 + 3\sqrt{7})^{20}(8 - 3\sqrt{7})^{20} = 1$

Sol.63. (b)

$7^n - 6n$

$(6 + 1)^n - 6n$

$[{}^nC_0 6^n + {}^nC_1 6^{n-1} + {}^nC_2 6^{n-2} + \ldots + {}^nC_{n-2} 6^2 + {}^nC_{n-1} 6^1 + {}^nC_n] - 6n$

$= [{}^nC_0 6^n + {}^nC_1 6^{n-1} + {}^nC_2 6^{n-2} + \ldots + {}^nC_{n-2} 6^2 + 6n + 1] - 6n$

$= [{}^nC_0 6^n + {}^nC_1 6^{n-1} + {}^nC_2 6^{n-2} + \ldots + {}^nC_{n-2} 6^2 + 1]$

$= 36[{}^nC_0 6^{n-2} + {}^nC_1 6^{n-3} + \ldots + {}^nC_{n-2}] + 1$

by dividing 36 there is 1 remainder.

Sol.64. (c)

$(1 + x)^p(1 + x)^q = (1 + x)^{p+q}$

Let x^3 exists its $(r + 1)^{th}$ term

that is General term,

$T_{r+1} = {}^nC_r (a)^{n-r} (b)^r$ in $(a+b)^n$

$= {}^{p+q}C_r x^r$

for coeff. of x^3, r = 3

so coeff. of x^3

${}^{p+q}C_3 = 35$

p + q = 7

Sol.65. (b)

General term of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{12}$ is

${}^{12}C_r \left(3^{\frac{1}{2}}\right)^{12-r}\left(5^{\frac{1}{4}}\right)^r$

By above expression, if r is multiple of 4, then there will be no rational term.

So r = 0, 4, 8, 12 i.e. 1^{st}, 5^{th}, 9^{th}, 13^{th} terms are rational terms.

Sol.66. (c)

Sum of binomial coefficients $= 2^n = 256$

So n = 8

Greatest term lies at middle position, and its middle term is $\left(\frac{8}{2}+1\right)^{th} = 5^{th}$ term

Sol.67. (b)

$k < \left(\sqrt{2}+1\right)^3 < k + 2$

$k < 2\sqrt{2} + 6 + 3\sqrt{2} + 1 < k + 2$

$k < 5\sqrt{2} + 7 < k + 2$

$k < 14.07 < k + 2$

k < 14.07 and k >12.07 so k = 13

Sol.68. (c)

$5^{99} = 5 \times 5^{98} = 5 \times 25^{49} = 5 \times (26 - 1)^{49}$

Each term of expansion of $(26 - 1)^{49}$ contains 26, except only last term ${}^{49}C_{49}(-1)^{49}$,

So last term will give remainder – 1 , and 5 is given in multiply so remainder will be – 5.

But remainder cannot be negative, when remainder comes negative than we add divisor in remainder so correct remainder is – 5 + 13 = 8

Sol.69. (d)

Greatest coefficient exists at middle term, and middle term of expansion of $(a + b)^{100}$ is 51^{th} term and this term is

$T_{51} = {}^{100}C_{50}(x)^{50}$ for $(1 + x)^{100}$

You can access the solutions of PYQ through 5 Youtube video:

1. Binomial theorem 2011 2012 2013
2. Binomial theorem 2014 2015
3. Binomial theorem 2016 2017 2018
4. Binomial theorem 2019 2020 2021
5. Binomial theorem 2022 2023 2024